

Reetesh MUKUL

Senior Principal Applied Scientist | Computer Vision and Multimodal AI

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Bangalore



Senior Principal Applied Scientist specializing in computer vision, large multimodal models, document AI, and video understanding. I build and lead research initiatives that connect model architecture, grounding, temporal reasoning, and production deployment. My work spans deep learning, generative AI, Bayesian reasoning, performance optimization, and scalable applied ML systems.

SKILLS

Research Areas	Deep Learning, Computer Vision, Large Multimodal Models, Vision Transformers, Diffusion Models, Generative AI, Document AI, Video Understanding, Bayesian Reasoning, Reinforcement Learning
Engineering	Distributed Systems, Parallel Programming, Performance Optimization, Functional Programming, Statistical Modeling, Advanced Mathematics, Classical Machine Learning
Programming Languages	C++20, Python, Haskell, Rust
Frameworks and Tools	PyTorch, OpenCV, TIMM, Pyro, scikit-learn, Boost, ONNX, Core ML, Git, Visual Studio Code, Xcode

PROFESSIONAL EXPERIENCE

Present	Senior Principal Applied Scientist, ORACLE INDIA, Bangalore
Dec 2023	Leading research and development for large multimodal models with a focus on document AI, video temporal reasoning, and grounding. My work includes model training, architecture design, and adapting multimodal systems for applied product use cases. Large Multimodal ModelsTransformersDocument AIVideo UnderstandingGrounding
Dec 2023	Principal Machine Learning Engineer, SINGAPORE RESEARCH CENTER, HUAWEI, Singapore
Dec 2022	Conducted and coordinated research initiatives for ad creatives, focusing on visual content understanding and the cognitive interaction between ad elements and users. Built deep learning vision models and text-attention models for advertisement analysis, with work across neural architectures, image attention, transformers, saliency, memorability, and Bayesian reasoning. PyTorchBayesian ReasoningVision TransformersCLIPDenoising Diffusion

Nov 2022 May 2021	Principal Machine Learning Engineer, INDIA RESEARCH CENTER, HUAWEI, India - Developed deep learning systems for automatic advertisement creation and creative ranking across industry, product, market, user, and artistic signals. - Modeled the causal contribution of individual creative components to overall advertisement performance. - Built CNN- and transformer-based tools for visual analysis, cognitive analysis, saliency prediction, blur prediction, memorability prediction, and graph-based component reasoning. - Developed NLP and sequential models to measure the impact of text in advertisement performance. PyTorch Advertisement Ranking Vision Transformers Memorability Cognition Denoising Diffusion
Apr 2021 Apr 2020	Machine Learning Developer, ADOBE, India - Developed and deployed machine learning models for Adobe Sensei. - Built models for visual similarity and post-processing components for panoptic segmentation. - Designed evaluation metrics for machine learning models. - Developed a tensor library supporting a subset of PyTorch functionality. PyTorch ONNX Core ML Keras TensorFlow CNN LSTM Clustering
Mar 2020 Sep 2016	Senior Software Developer Machine Learning Engineer, ADOBE, India - Improved performance for Lightroom Classic import, grid, and library workflows. - Designed a Markov Decision Process-based dynamic import algorithm for distributed batch allocation. - Built a lightweight embedded performance database to profile Lua code automatically. - Used common table expressions to deliver low-latency queries for folders and keywords. - Developed natural language models for photographic feature prediction, user analytics, churn analysis, and ML-assisted resource management. Xcode Visual Studio Lua SQLite R Python Keras PyTorch Hadoop Hive LSTM Reinforcement Learning Distributed Programming Scheduling Markov Decision Processes
Aug 2016 Aug 2015	Senior Software Developer Machine Learning Engineer, FLIPKART, India - Developed machine learning models for ad click prediction, including CTR, CVR, ad ranking, and sampling. - Contributed to the mobile ad SDK for native advertising on Android. Ads Native Ads Pandas R Ranking Sampling Android Thompson Sampling Distributed Programming
Jul 2015 Dec 2011	Senior Software Developer Machine Learning Engineer, QUALCOMM, India - Developed optical character recognition systems for Indic Devanagari characters. - Built online detection capabilities for interesting scenes in camera workflows. Computer Vision C++ OpenCV Android Python Assembly Parallel Programming
Nov 2011 June 2008	Software Developer, QUALCOMM, India Developed voice state machine components and embedded drivers. C++ Embedded Systems Parallel Programming

May 2008
June 2006

Software Developer, TEXAS INSTRUMENTS, India

➤ Developed software for Global Positioning System components.

C++ Embedded Systems Assembly Language



PROJECTS

CREATIVE ADVERTISEMENT (HUAWEI)

2021 - PRESENT

Automated advertisement creation informed by composition and consumer psychology

Developed systems for automated advertisement creation by modeling market, industry, product, user, and artistic signals. The work focused on generating creatives that influence user emotion and perception while improving product and brand outcomes.

PyTorch Cognitive Science Bayesian Reasoning Computer Vision

LRSENSEI (ADOBE)

2020 - 2021

Computer vision for best-photo selection, auto-stacking, and panoptic segmentation

Worked on Best Photos, visual similarity, and panoptic segmentation for Lightroom Sensei. Improved model quality, defined evaluation metrics, clarified user requirements, and evaluated deployment options across client-side and server-side execution with CPU and GPU scheduling constraints.

Torch ONNX CoreML Feature Pyramids Contrastive Learning

LRPERFORMANCE (ADOBE)

2016 - 2020

Performance optimization and user analysis for Lightroom Classic

Improved Lightroom performance through efficient algorithms, deep parallelism, resource management, and machine learning. Used common table expressions and system-level parallelism to reduce latency, then applied reinforcement learning, Markov Decision Processes, and bandit methods for resource estimation. Also used user logs to predict feature adoption, churn, and conversion likelihood.

C++ Lua SQLite Markov Decision Processes Reinforcement Learning Keras LSTM Hive Hadoop

DEVANAGARI OCR (QUALCOMM)

2011 - 2015

Optical Character Recognition for Devanagari

Developed an optical character recognition system for Indic Devanagari characters, including both character decoders and word decoders. Optimized the system for mobile-device constraints through cache-aware vector routines, Chandrabindu detection, and word-decoder modules.

C++ Computer Vision Machine Learning Android



PAPERS AND PATENTS

1. Photo-Editing Application, Patent No. 10,884,769
2. Markov Decision Process for Efficient Data Transfer
3. Context-based Recommendation System for Feature Search
4. Environment-Aware Application-Based Resource Management Using Reinforcement Learning
5. Automatic Teeth Whitening Using Teeth Region Detection and Individual Tooth Location, U.S. Patent Application 20200342586
6. Photo-Editing Application Recommendations
7. Parameter Estimation for Accelerometers, Processes, Circuits, Devices, and Systems. US20090259424A1

8. Broadband HF/VHF/UHF Communication on Power Lines. WO2007000777A1.
9. R. Mukul et al., "An adaptive bandwidth request mechanism for QoS enhancement in WiMAX real-time communication," 2006 IFIP International Conference on Wireless and Optical Communications Networks, Bangalore, 2006, pp. 5, doi: 10.1109/WOCN.2006.1666583.



EDUCATION

2006 **M.Tech**, IIIT Bangalore

2004 **B.Tech**, BIT Sindri



LANGUAGES

English



Hindi



STRENGTHS

- Continuous learner
- Research-driven
- Self-motivated
- Autonomous